

MAYANK BANSAL

Machine Learning Engineer

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

TECHNICAL SKILLS

Machine Learning and AI: Scikit-learn, CNNs, Neural Networks, Model Evaluation, Feature Engineering, Exploratory Data Analysis, Retrieval Augmented Generation

Deep Learning Frameworks: TensorFlow, PyTorch

Data and Visualization: NumPy, Pandas, Matplotlib, Seaborn

Backend and Deployment: Flask, REST APIs, Docker

Programming Languages: Python, JavaScript, Java, C++, SQL

PROJECTS

AI Complaint Triage & Routing System | [GitHub](#)

Jan 2026 – Feb 2026

- **Objective:** Automate complaint handling at scale by building an end-to-end ML triage system routing across 8 departments with urgency prediction and confidence aware outputs
- **Approach:** Designed a hybrid credibility scoring pipeline combining rule-based logic and ML confidence to assess actionability and trigger follow ups
- **System Design:** Engineered a modular Flask REST API with separated inference, credibility logic, and explainable decision flow. Designed and published an 800-sample dataset for multi-class complaint routing and urgency classification on Kaggle.
- **Outcome:** Enabled auditable, feedback driven complaint handling with a foundation for continuous model improvement by retraining

Tech: Python, Flask, Scikit-learn, TF-IDF, SQLite

PDF Question Answering System (RAG) | [GitHub](#)

Dec 2025 – Jan 2026

- **Objective:** Build a reliable question answering system over unstructured PDFs by grounding LLM responses in retrieved document context
- **Approach:** Implemented a RAG pipeline using document chunking with overlap, FAISS vector similarity search, and Sentence Transformer embeddings
- **System Design:** Engineered prompt level guardrails to reduce hallucinations and enforce explicit 'I don't know' responses for out-of-scope queries
- **Outcome:** Delivered transparent, verifiable answers with source attribution via a Flask backend and HTML, CSS, JavaScript frontend

Tech: FAISS, Sentence Transformers, Flask, Gemini API

Visual Search Engine | [GitHub](#)

Feb 2026 – Mar 2026

- **Objective:** Build an image-based retrieval system that enables users to search for visually similar images using deep learning embeddings instead of text queries
- **Approach:** Utilized a pretrained ResNet50 model as a feature extractor by removing the classification layer to generate 2048-dimensional embeddings, and implemented **FAISS** for efficient nearest-neighbour similarity search
- **System Design:** Engineered a FastAPI-based REST backend integrating image preprocessing, embedding generation, and vector search, along with a responsive frontend for image upload, preview, and dynamic result display
- **Outcome:** Delivered real-time visual search over a fashion dataset with fast and accurate similarity retrieval, demonstrating end-to-end integration of computer vision, vector search, and full-stack deployment

Tech: PyTorch, FAISS, FastAPI, NumPy, JavaScript, HTML, CSS

Mood2Mail – Real-Time Email Tone Analyzer | [GitHub](#)

Apr 2025 – Apr 2025

- **Objective:** Help users improve professional communication by detecting and correcting unintended email tone before sending
- **Approach:** Built an NLP based tone classifier using TF-IDF vectorization and **Multinomial Naive Bayes** across five tone categories
- **System Design:** Created and manually annotated a 475-sample real world email dataset and validated performance through preprocessing and train test evaluation, also published the same on Kaggle.
- **Outcome:** Delivered real-time tone feedback through a responsive Flask and JavaScript web interface with 96.84% accuracy (weighted F1: 0.97 across 5 tone categories), enabling proactive tone adjustment

Tech: TF-IDF, Naive Bayes, Flask, JavaScript

ACHIEVEMENTS

- Published a research paper, *Mood2Mail: A Lightweight Real Time Email Tone Analyzer Using Classical Machine Learning*, on Zenodo with DOI, covering dataset creation, TF IDF feature engineering, and model evaluation
- Solved 580+ competitive programming problems on GeeksForGeeks with a 101-day coding streak and a rank of 159 among 39,000+ participants along with 100+ problems on LeetCode

CERTIFICATIONS

- Complete Data Science and Machine Learning, GeeksForGeeks Jun 2025 – Nov 2025
- AI Tools Skill Up, GeeksForGeeks Jun 2025 – Aug 2025

EDUCATION

Lovely Professional University – Bachelor of Technology in Computer Science Engineering

Phagwara, Punjab, India | Aug 2024 – May 2028 | **CGPA: 9.33/10**

St. Joseph's International School – XII Senior Secondary (Physics, Chemistry, Mathematics)

Hisar, Haryana, India | Apr 2023 – Apr 2024